BRANDON SANTOR

GEOSPATIAL RESEARCH | SCIENTIFIC SOFTWARE | U.S. MARINE CORPS VETERAN

brandon.santor@tamu.edu | emp1241.github.io

Geospatial researcher and software developer who leads research teams and builds the tools they use. Independently created software for tree-ring and hydrothermal-vent research; lead NWSW research, COLE development, and a five-person Mars Explorer team. Build tools that prepare research data, connect maps with their sources, and turn elevation measurements into explorable 3D terrain. Texas A&M geographic information science undergraduate and U.S. Marine Corps veteran.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant & Software Developer | Torbenson Lab

March 2026-present | Paid appointment since June 2026

- Independently built the Tree-Ring Atlas Explorer (TRAE) in response to lab needs; its research use led to a paid appointment for continued software development and research support.
- Added reviewed data imports, linked maps and graphs, climate matching, saved work sessions, and exports that record source and method details. Built tools for PRISM gridded climate data, NetCDF multidimensional files, ROXAS wood-anatomy outputs, and year-labeled WalDendro X-ray records.
- Used TRAE to derive summaries, organize study data, and prepare exports for a co-authored manuscript in preparation; its screening summaries support analysis but do not replace scientific validation.

Research Lead & GIS Developer | New World Screwworm (NWSW)

Fall 2026-present | Texas A&M GIS Practicum | Client: U.S. Department of Agriculture (USDA)

- Lead student NWSW research for USDA and reconcile official records; recovered 2,629 missing outbreak-start dates through exact record matching, restoring timing information for chronological comparisons.
- Connected livestock figures to places, rebuilt transport records, checked weather estimates, and made the map-building process repeatable. Tested uncertain report locations and used 64 million random trials to establish how long the calculation needed to run to keep random sampling noise below the study's target.
- Used my Research Architect framework to build the team's coordination system and independently developed the Spatial Evidence Workbench, keeping sources, maps, review decisions, and shared files connected.

Independent Esri GIS Technical Consultant & Instructor

February-May 2026

- Audited seven graduate GIS modules and rebuilt six workflows into 189 pages of illustrated technical manuals.
- Corrected GIS software and projection problems, misaligned map grids, database structure and geographic-join errors, licensing issues, and desktop-to-web failures.
- Built checks into each workflow and explained the steps through illustrated manuals and instruction, enabling the client to run the corrected analyses independently.

High School Baseball Coach

2013-2020 | Paid program coach

- Coached nearly 100 student-athletes each year, developing baseball skills, teamwork, and discipline.
- Coordinated conditioning, equipment, and maintenance for the program.
- Mentored student-athletes and emphasized education throughout their high school experience.

U.S. Marine Corps | Infantryman, Acting Squad Leader & Logistics NCO

2011-2013

- Managed weapons, ammunition, medical supplies, and safety equipment supporting approximately 350 Marines.
- Served as acting squad leader for 13 Marines.
- Prepared Marines for combat through live-fire training.

TEKsystems | Network & Computer Systems Contractor

2007-2011

- Executed enterprise hardware and software deployments involving 100+ devices per project during nationwide infrastructure upgrades.
- Independently diagnosed complex computer and network systems issues across changing client environments.
- Installed, configured, and supported systems while coordinating with local and remote technical teams.

SELECTED SOFTWARE & RESEARCH

New World Screwworm (NWSW) Research

Research lead & GIS developer | Texas A&M GIS Practicum | USDA client

- Lead the research team investigating reported screwworm detections and their uncertain locations. Recovered 2,629 missing outbreak-start dates, reconciled livestock and transportation evidence, inspected satellite imagery, and built repeatable maps and location-simulation experiments. Developed the team's research infrastructure to preserve and review that work.

Mars Explorer

Team lead | Database & 3D systems

- Lead a five-person team; developed Mars coordinate handling, image-to-elevation alignment, the spatial database, and the 3D terrain pipeline. Released version 1.0.0 for browser and Windows, connecting four Curiosity areas, mapped rover stations, panoramas, source-linked science, and a 200-image gallery.

Thermal Vent Atlas Explorer

Independent creator & developer

- Built and delivered TVAE to Sylvan Lab for review and internal use, mapping 721 preserved InterRidge vent-field records with seafloor relief, tectonic boundaries, local research fields, comparisons, and saved sessions.

COLE | Cost Of Living Explorer

Application creator | Developer | Project lead

- Created and lead COLE; corrected and integrated contributor additions into five feature modules, led outdoor-space data repair across 3,143 counties, and released version 1.0.1 for web and Windows.

Research Architect (RA)

Creator & principal designer | PhalAInx independent software studio

- Built a Python framework that gives each team member's AI assistant its role, methods, and reviewed sources. Added workspace checks, plan approval, saved decisions, and recovery tools; used it to establish the NWSW team's coordination system.

High Command (HC)

Creator & developer | PhalAInx independent software studio

- Built a Windows workspace for projects and AI conversations using Tauri, React, Rust, and SQLite. Added resumable imports and checks that prevent old edits from overwriting newer work. Built and live-tested separate Codex and Antigravity connections; a Windows candidate verifies installed Codex clients and stops commands during storage stalls. Provider integration remains in development.

The Deception Gap | Smartphone GNSS

Independent continued research

- Expanded a class project into independent satellite-positioning research, more than doubling the original collection to 781 retained readings from three smartphones across five crowd categories; led continued field collection, statistical analysis, interpretation, and poster development.

NASA SUITS Challenge Proposal

Team lead & primary system designer | 2021-2022

- Led a student team's HoloLens 2 interface proposal for extravehicular activity (EVA), connecting local positioning, terrain overlays, and a command hub intended to distribute current crew positions and shared pins; co-authored technical documentation submitted to NASA for review.

EDUCATION

Texas A&M University

Expected May 2027 | B.S. in Geographic Information Science & Technology | Institutional GPA: 4.0 / 4.0

A.A. Liberal Arts

Blinn College | 2024

A.S. Computer Networking

ITT Technical Institute | 2007

HONORS & SCHOLARSHIPS

Second Place | TxGIS Day Undergraduate Student Research Presentation Competition | 2025 | Health Geography GIS and statistical analysis

Galloway '29 Aggie Veteran Freedom Scholarship II | Two-time recipient | 2025-2027

Dennis W. Hughes '79 & Melinda Hood Hughes '81 Endowed Scholarship in Geography | Two-time recipient | 2025-2027

Richard A. & Mollie B. Williford Dean's Scholarship | College of Arts & Sciences | 2026-2027

TECHNICAL TOOLKIT

Languages: Python; R; JavaScript / TypeScript; HTML / CSS; Rust; SQL.

Geospatial tools: ArcGIS Pro / Field Maps; QGIS / GDAL; GeoPandas / Shapely / PyProj / Pyogrio; PostgreSQL / PostGIS; Cesium / Leaflet; Rasterio.

Application development: FastAPI / Flask / Waitress; React / Vite / Tauri 2; Node.js / Caddy; SQLite; PyInstaller / Windows packaging; JSON Schema.

Research methods & data: NumPy / pandas; NetCDF / PRISM / RWL / CRN; Monte Carlo precision checks; Smartphone satellite-positioning field analysis; Research source tracking; SHA-256 manifests.

RESEARCH COMMUNICATION

Explain technical methods through illustrated manuals, research presentations, and a co-authored manuscript in preparation with Torbenson Lab.